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METHODIST UNIVERSITY GHANA

**PHYSICS FOR HEALTH SCIENCES
END-OF-SECOND SEMESTER EXAMINATION**

TIME ALLOWED: 1 HOUR: 40 MINUTES

**ANSWER ALL QUESTIONS ON SECTION A AND ANY OTHER TWO QUESTIONS
FROM SECTION B INCLUDING QUESTION ONE
LEVELS 100 AND 200
DANSOMAN AND TEMA CAMPUS**

SECTION A

Answer all questions

- 1) A concerned citizen contacts the local environmental agency to inquire about the regulation of ionizing radiation from human activities. Which governmental agency is responsible for setting protective limits on such radiation?
 - A. Food and Drug Administration (FDA)
 - B. Environmental Protection Agency (EPA)
 - C. World Health Organization (WHO)
 - D. International Atomic Energy Agency (IAEA)

- 2) A patient expresses concern about potential health risks associated with living near power lines. Which type of radiation emitted by power lines is considered non-ionizing?
 - A. X-rays
 - B. Gamma rays
 - C. Alpha particles
 - D. Electromagnetic fields

- 3) A patient presents to the emergency department with a suspected internal contamination from a radioactive source. Which type of radiation, known for its high ionization potential and ability to cause severe damage if ingested, should the medical team be particularly concerned about?
 - A. Alpha radiation
 - B. Beta radiation
 - C. Gamma radiation
 - D. Neutron radiation

- 4) A radiologist is reviewing a nuclear medicine scan of a patient with suspected bone metastases. Which radiotracer, commonly used in nuclear medicine for bone imaging, emits gamma radiation?
 - A. Technetium-99m
 - B. Iodine-131
 - C. Fluorine-18
 - D. Carbon-14

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- 5) Which of the following outcomes may result from cells experiencing irreparable DNA damage due to radiation exposure?
 - A. Programmed cell death
 - B. Nonlethal DNA mutation
 - C. Premature aging and cancer
 - D. Repair of the damage
- 6) What term describes the process by which a cell experiences DNA damage and undergoes programmed cell death, eliminating potential genetic damage from the larger tissue?
 - A. Apoptosis
 - B. Mutation
 - C. Ionization
 - D. Absorption
- 7) Which of the following is NOT a somatic effect of chronic radiation exposure?
 - A. Premature aging
 - B. Skin burns
 - C. Cancer
 - D. Genetic mutations
- 8) Which statement accurately describes the relationship between radiation dose and biological damage?
 - A. The higher the dose rate, the smaller the biological damage.
 - B. Radiation dose has no impact on biological damage.
 - C. The higher the dose rate, the larger the biological damage.
 - D. Biological damage remains constant regardless of dose rate.
- 9) What is the recommended method for disposing of patient waste or excreta containing radioactive material?
 - A. Flushing down the drain
 - B. Burning in an incinerator
 - C. Collecting and disposing with other radioactive waste
 - D. Burying in a backyard landfill
- 10) What is the primary focus of the initiative on radiation safety in healthcare settings?
 - A. Risk assessment
 - B. Risk management
 - C. Risk communication
 - D. All of the above
- 11) Which of the following best defines radiation?
 - A. The emission of light from a heated object
 - B. Any process in which energy travels through a medium or through space
 - C. The process of radioactive decay
 - D. The generation of heat from nuclear reactions

- 12) Which organization is primarily concerned with safety relating to exposure to ionizing radiation?
 - A. WHO (World Health Organization)
 - B. IAEA (International Atomic Energy Agency)
 - C. CDC (Centers for Disease Control and Prevention)
 - D. EPA (Environmental Protection Agency)

- 13) Which of the following is NOT a source of ionizing radiation?
 - A. X-ray machines
 - B. Cell phones
 - C. Isotopes of uranium
 - D. Cosmic rays

- 14) A radiology technician is preparing to perform an industrial sterilization process using radiation. Which of the following radionuclides is commonly used for this purpose?
 - A. Uranium-238
 - B. Potassium-40
 - C. Cobalt-60
 - D. Technetium-99m

- 15) Which type of radiation exposure is associated with the occurrence of various types of skin cancer, particularly from sunlight exposure?
 - A. Chronic effects
 - B. Acute effects
 - C. Non-ionizing radiation
 - D. Ionizing radiation

- 16) Which type of radiation is associated with the risk of photochemical cataracts in the eye and erythema (skin reddening)?
 - A. Visible light
 - B. Infrared radiation
 - C. Microwaves
 - D. Radiofrequency radiation

- 17) Which type of radiation is commonly associated with remote controls and heating of the body surface?
 - A. Visible light
 - B. Infrared radiation
 - C. Microwaves
 - D. Radiofrequency radiation

- 18) What are the three main ways people are exposed to radiation?
 - A. From external sources, ingestion of radionuclides, absorption through the skin
 - B. From the atmosphere, through water consumption, internal inhalation
 - C. From sunlight, microwave radiation, radiofrequency exposure
 - D. From cellular devices, medical procedures, environmental contamination

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- 19) What is the primary purpose of the ALARA principle in radiation safety?
- A. To maximize radiation exposure levels
 - B. To minimize radiation exposure levels
 - C. To standardize radiation measurement units
 - D. To regulate the use of radioactive materials
- 20) What is the primary focus of the initiative on radiation safety in healthcare settings?
- A. Risk assessment
 - B. Risk management
 - C. Risk communication
 - D. All of the above
- 21) Which of the following is an example of a scalar quantity?
- A. Velocity
 - B. Force
 - C. Time
 - D. Acceleration
- 22) Which of the following is a basic physical quantity?
- A. Area
 - B. Speed
 - C. Density
 - D. Time
- 23) What type of physical quantity is mass?
- A. Scalar quantity
 - B. Derived quantity
 - C. Vector quantity
 - D. Basic quantity
- 24) Which of the following is an example of a derived physical quantity?
- A. Length
 - B. Mass
 - C. Speed
 - D. Time

Use the statement below to answer questions 25 and 26.

Suppose a car starts from rest and accelerates uniformly to a speed of 5m/s in 10s.

- 25) Find acceleration
- A. 0.5 m/s^2
 - B. 25 m/s^2
 - C. 2.5 m/s^2
 - D. 0.25 m/s^2

26) The distance moved in this time

- A. 0.5
- B. 25 m
- C. 2.5 m
- D. 0.05 m

Use the statement below to answer questions 27 and 28.

Suppose a car travelling at 5m/s is brought to rest in a distance of 20m.

27) Find its deceleration

- A. -0.625m/s^2
- B. -0.625m/s^2
- C. -0.625m/s^2
- D. -0.625m/s^2

28) The time taken to stop

- A. 8 sec
- B. 0.8 sec
- C. 0.008 sec
- D. 80 sec

Use the statement below to answer questions 29 and 30.

A car starts from rest and accelerates at 4m/s^2 through a distance of 20m.

29) How fast is it then going?

- A. 1.265 m/s
- B. 1265 m/s
- C. 12.65 m/s
- D. 0.1265 m/s

30) How long did it take?

- A. 316 sec
- B. 3.16 sec
- C. 0.316
- D. None of the above

31) Find the time taken for a car to travel 98m if it starts from rest and accelerates at 4m/s^2 .

- A. 7 sec
- B. 0.7 sec
- C. 70 sec
- D. None of the above

32) A car is moving at 72km/h when it begins to slow down from point 'A' with a deceleration of 1.5m/s^2 . How long does it take to travel 70m to point 'B' as it slows down?

- A. 41.4 sec
- B. 414 sec
- C. 4.14 sec
- D. None of the above

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- 33) The ball bounces back on hitting the ground. A ball strikes the ground with certain force (Action) and the ground pushes back the ball with equal force (Reaction)---is an example of which law of motion?
- A. first law
 - B. second law
 - C. third law
 - D. all of these
- 34) In an inertial frame of reference, the vector sum of the forces F on an object is equal to the mass m of that object multiplied by the acceleration a of the object. What is the name of this law of motion?
- A. first law
 - B. second law
 - C. third law
 - D. all of these
- 35) Which of Newton's Laws gives the reason for why you can feel things that you touch?
- A. first law
 - B. second law
 - C. third law
 - D. all of these
- 36) How much work is done by a 10 N force pushing a box 4 m across the floor?
- A. 40 J
 - B. 14 J
 - C. 2.5 J
 - D. 0.4 J
- 37) If a motor does 500 J of work to lift a 200 N weight, how high was the weight lifted?
- A. 0.4 m
 - B. 2.5 m
 - C. 300 m
 - D. 0.3 m
- 38) A nurse dryer transforms 10,000 J of energy in 4 s. What is its power?
- A. 40,000 W
 - B. 2,500 W
 - C. 25,000 W
 - D. 250 W
- 39) A 60 W lamp is left on for 20 s. How much energy has been transformed?
- A. 1,200 J
 - B. 80 J
 - C. 3 J
 - D. 120 J
- 40) In which of the following states does a body possess kinetic energy?
- A. Rest
 - B. Motion
 - C. When placed on a platform
 - D. In zero gravity

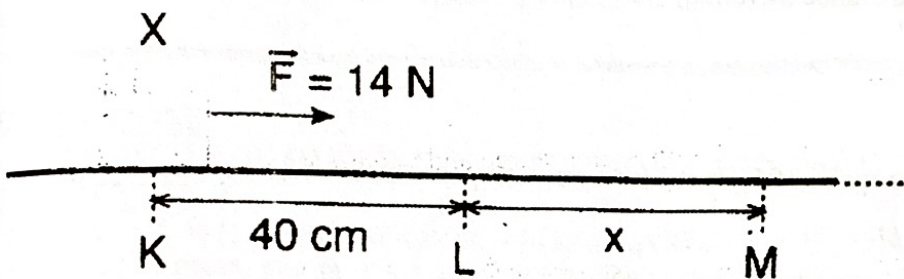
SECTION B

NB: You are to answer **two** questions in this section. **Question 1** and any other question.

QUESTION ONE

a. Use the statement below to answer the following questions i and ii

The friction force between KL is 9N and between LM is 5 N. If the work done between KL and LM are equal,



- i. Find the distance between LM
 - ii. Work done by 10N between KL is 8 joules. Then, what is the distance x?
- b. If the work done to 2 kg object is 6.4 j at a distance of 70cm. What is the force of friction?
 - c. A horse is pulling a cart with a force of 600 N by 28m is 60 s, calculate the power at which the horse is pulling the cart.
 - d. A man lifts a weight of 600N through a vertical line with an average velocity of 2m/s. What is the power of the man?
 - e. A(n) _____ is a push or a pull.
 - f. State two uses of Friction
 - g. State three disadvantages of friction

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QUESTION TWO

A bus travelling in a straight line with uniform acceleration has an instant speed of 6 m/s . 5 seconds later, has reached a speed of 36 m/s .

Find

- the magnitude of its acceleration
- the distance travelled in 5 seconds?
- the total distance travelled

QUESTION THREE

A bullet moving with the speed of 5 m/s strikes a tree and penetrates 3.5 cm before stopping.

Find

- the magnitude of the acceleration and
- time taken to stop?